CISS245: Advanced Programming
Quiz q6501

Name: eagraham1@cougars.ccis.edu Score:

Q1. A matrix is a 2D array of integers (for this question). Here's how you use it:

```
Mat m(2, 3); // m models a 2-by-3 array of integers, all initialized to 0
std::cout << m << '\n'; // prints like this:
                        // 0 0 0
                        // 0 0 0
m(0, 0) = 42;           // sets row 0, column 0 entry to 42
std::cout << m << '\n'; // prints like this:
                        // 42 0 0
                        // 0 0 0
Mat n(2, 2);            // n is 2-by-2
n = m;                  // n is 2-by-3 (like m) and values of m are copied over
std::cout << n << '\n'; // prints like this:
                        // 42 0 0
                        // 0 0 0
Mat p(m);
std::cout << p << '\n'; // prints like this:
                        // 42 0 0
                        // 0 0 0
```

Complete the following header file. Do not implement any methods/functions in the header file (that's done in the question below).
Answer:

```
// file: Mat.h
#include <iostream>
class Mat
{
public:
Mat(int r, int c);

Mat(const Mat& other);
Mat & operator=(const Mat & other);

~Mat();

int & operator()(int i, int j);
const int & operator()(int i, int j) const;
private:
    int num_rows_; // number of rows
    int num_cols_; // number of columns
```

```
    int * p_;      // there's only one reason for this!
};
std::ostream & operator<<(std::ostream &, const Mat &);
```

Q2. Refer to the question above. Complete the following implementation file for Matrix class. Obviously there must be no memory issues.
Answer:

```
// file: Mat.cpp
#include <iostream>
#include "Mat.h"

Mat::Mat(int r, int c)
: num_rows_(r), num_cols_(c), p_(new int[r * c])
{
  for (int i = 0; i < r * c; ++i)
  {
    p_[i] = 0;
  }
}

Mat::Mat(const Mat & other)
: num_rows_(other.num_rows_), num_cols_(other.num_cols_),
p_(new int[other.num_rows_ * other.num_cols_])
{
  for (int i = 0; i < num_rows_ * num_cols_; ++i) {
    p_[i] = other.p_[i];
  }
}

Mat & Mat::operator=(const Mat & other)
{
  if (this != &other)
  {
    delete[] p_;

    num_rows_ = other.num_rows_;
    num_cols_ = other.num_cols_;
    p_ = new int[num_rows_ * num_cols_];

    for (int i = 0; i < num_rows_ * num_cols_; ++i)
    {
      p_[i] = other.p_[i];
    }
  }
  return *this;
}

Mat::~Mat()
{
```

```
  delete[] p_;
}

int & Mat::operator()(int i, int j)
{
  return p_[i * num_cols_ + j];
}

const int & Mat::operator()(int i, int j) const
{
  return p_[i * num_cols_ + j];
}

std::ostream & operator<<(std::ostream & cout, const Mat & mat)
{
  for (int i = 0; i < mat.num_rows_; i++)
  {
    for (int j = 0; j < mat.num_cols_; j++)
    {
      cout << mat(i, j);
      if (j < mat.num_cols_ - 1)
      cout << ' ';
    }
    cout << '\n';
  }
  return cout;
}
```

You are advised to use google's asan tool.

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.